



International Journal of Sports Physiology and Performance

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Additional Benefits of Repeated-Sprint Training With Prolonged End-Expiratory Breath Holding for Improving Repeated-Sprint Ability in Semiprofessional Soccer Players

in [International Journal of Sports Physiology and Performance](#)

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DOI: <https://doi.org/10.1123/ijsp.2024-0444>

Keywords: [hypoventilation](#); [hypoxia](#); [hypercapnia](#); [team sports](#); [high-intensity exercise](#)

First Published Online: 24 Feb 2025

In Print: [Volume 20: Issue 4](#)

Page Range: 568–574

Purpose: To investigate the effects of running repeated-sprint training with voluntary hypoventilation at low lung volume (RSH-VHL) including prolonged end-expiratory breath holding (EEBH) on running repeated-sprint ability (RSA). **Methods:** Twenty semiprofessional male soccer players completed 12 sessions of repeated 50-m running sprints over a 6-week period either with EEBH (RSH-VHL, $n = 10$) or with normal breathing (RSN, $n = 10$). Before (Pre) and after (Post) training, a running RSA test consisting of performing maximum 30-m “all-out” sprints until task failure, with a minimum of 10 sprints, was implemented. **Results:** The maximum number of sprints was increased at Post compared to Pre in RSH-VHL only (13.5 [4.4] vs 7.7 [2.3], $P < .01$) and was greater in RSH-VHL than in RSN at Post ($P < .01$). The mean velocity for sprints 1 to 10 and sprints 6 to 10 was increased in both groups at Post ($P < .01$) but was greater in RSH-VHL than in RSN after the training period ($P < .01$). There was no change in the reference velocity ($P = .80$) or in the maximal velocity reached during the RSA test ($P = .52$) in either group. The mean minimal

arterial oxygen saturation recorded during training at the end of the sprints was lower in RSH-VHL (78.5% [1.4%]) than in RSN (97.3% [0.1%]). **Conclusions:** This study shows that an RSH-VHL intervention including prolonged EEBH can provide an additional benefit for improving RSA in male semiprofessional soccer players. This result may be due in large part to the strong hypoxic effect induced by the prolonged EEBH.

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